

2011 Year 6 H1 Chemistry Preliminary Exams – Paper 2 Answers

- 1 Naturally occurring magnesium has the following isotopic abundances:

Isotope	Relative isotopic mass (amu)	Abundance(%)
^{24}Mg	23.98504	78.99
^{25}Mg	24.98584	10.00
^{26}Mg	25.98259	11.01

- (a) Define the term relative isotopic mass.

The relative isotopic mass of an element is the **mass of an atom of the isotope relative to 1/12 the mass** of an atom of ^{12}C .

- (b) Calculate the average atomic mass of Mg.

$$A_r \text{ of Mg} = 23.98504 \times 78.99/100 + 24.98584 \times 10/100 + 25.98259 \times 11.01/100 = \mathbf{24.31}$$

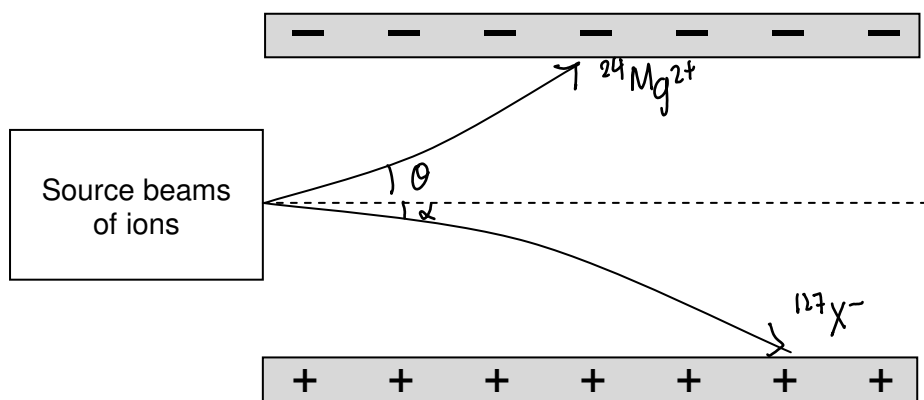
- (c) 0.02 g of ^{26}Mg isotope is allowed to decay. If the half life of magnesium isotope is 12.5 hours, what mass of isotope left after 25 hours?

0 $\rightarrow$ 12.5 hours $\rightarrow$ 25 hours (2 half lives)

Mass of isotope = **0.005g**

- (d) A source beam of $^{24}\text{Mg}^{2+}$ and $^{127}\text{X}^{-}$ is passed through an electric field. Sketch the deflection paths of the ions in the diagram below.

Show clearly their relative angles of deflection.



- Angle of deflection proportional to q/m
- $\delta < \theta$

- (e) Describe the structure of magnesium chloride and explain why magnesium chloride has a high boiling point.

Magnesium chloride has a **giant ionic lattice structure** where **oppositely charged ions** are held by **strong electrostatic forces of attraction** which requires **a large amount of energy** to break the strong ionic bonds.

- (f)(i) State the pH of the solution formed when MgCl_2 is dissolved in water.
6.5

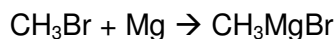
- (ii) Write equations to show the reaction of MgCl_2 with water.



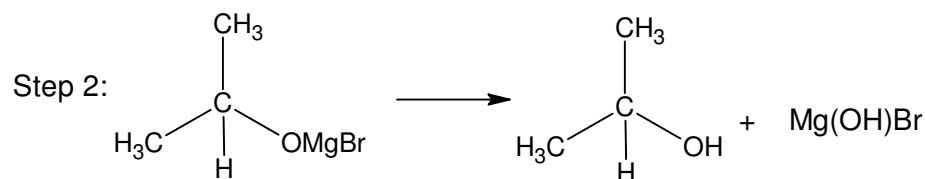
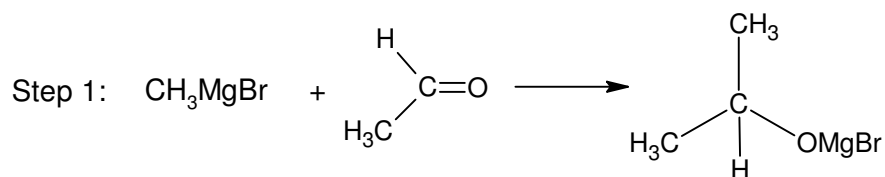
- (iii) Predict the expected observation when MgCl_2 is added to test-tube containing aqueous sodium carbonate.
Effervescence of carbon dioxide gas will be observed.

- (g) Magnesium metals can be used to make grignard reagents, which are alkyl- or aryl magnesium halides, often used for organic synthesis reactions.

In the preparation of a Grignard reagent, a halogenoalkane is added to magnesium solids in a flask containing dry ether as a solvent. The mixture is warmed over a water bath. The reaction is as follows:



This Grignard reagent can be used to synthesise propan-2-ol from ethanal. The reaction takes place in two steps as shown below:



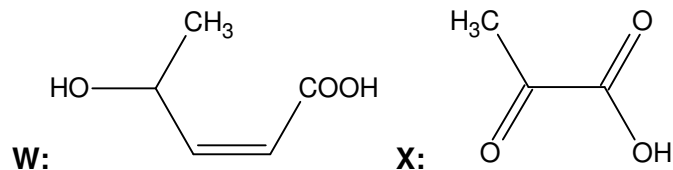
- (i) State the type of reaction taking place in Step 1.
(Nucleophilic) **Addition**
- (ii) Suggest a suitable carbonyl compound that can be used to synthesise butan-2-ol
using the Grignard reagent given.
 $\text{C}_2\text{H}_5\text{CHO}$

[Total: 15]

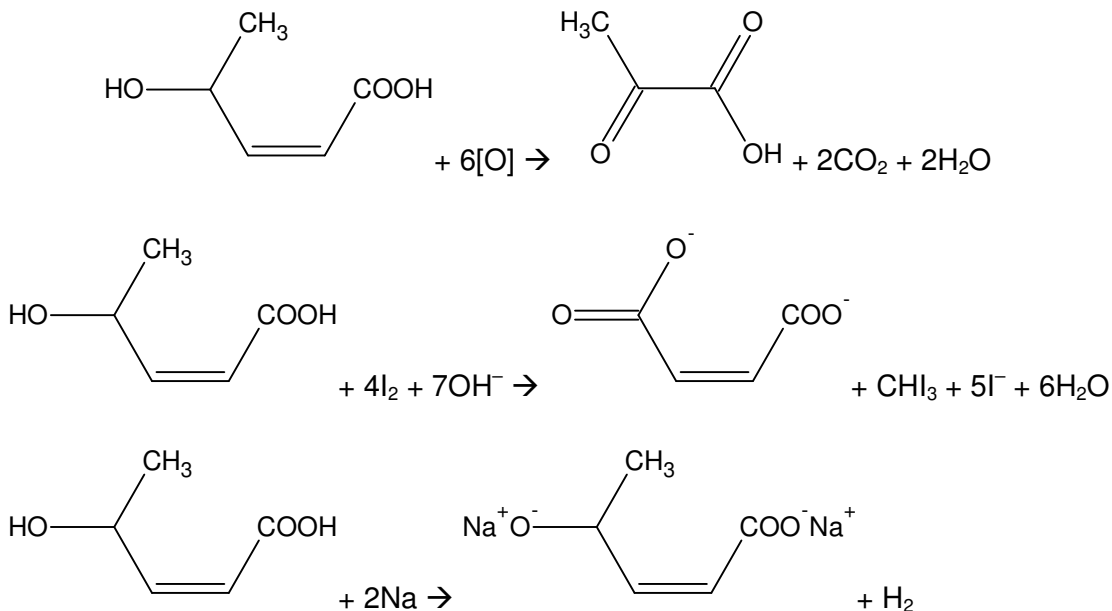
- 2 Compound **W** has the molecular formula $\text{C}_5\text{H}_8\text{O}_3$. When **W** is refluxed with acidified potassium manganate(VII), only one organic product, **X**, is obtained. When **W** is warmed with aqueous sodium hydroxide and iodine, a yellow precipitate is formed. **W** has no reaction when 2,4-dinitrophenylhydrazine is added. One mole of **W** liberates one mole of hydrogen gas when treated with sodium metal.

- (a) State the types of reaction undergone when **W** is
- (i) refluxed with acidified potassium manganate(VII); and
Oxidation
- (ii) warmed with aqueous sodium hydroxide and iodine.
Oxidation
- (b) Name all the possible functional groups present in **W**.
(Secondary) alcohol, alkene and carboxylic acid group. Ester

- (c) Suggest the two possible structures of **W** and hence identify the organic product **X**.

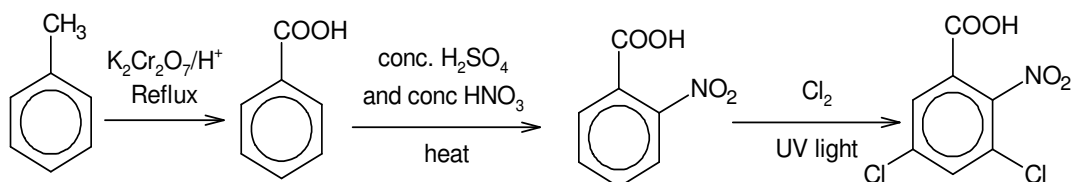


- (d) Write balanced equations for all the reactions which have occurred.



[3]

- (e) A Chemistry professor wanted his student to devise a synthetic method to convert methylbenzene to 2-nitro-3,5-dichloro acid. His student suggested this method:



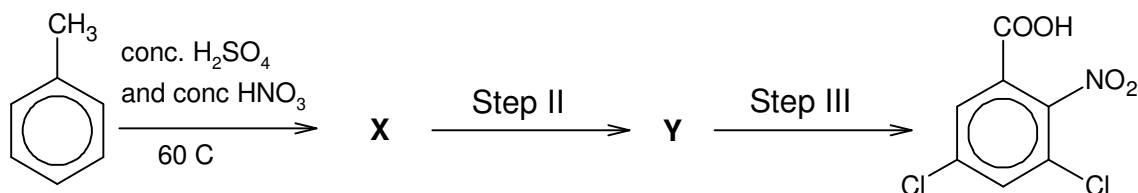
The professor identified at least three errors to his suggested method for the synthesis.

- (i) Identify the three errors and explain with reasons.

1) $\text{K}_2\text{Cr}_2\text{O}_7$ is too **mild an oxidising agent and cannot** to oxidize the methyl side chain. (Acidified KMnO_4 has to be used).

- 2) COOH is **3-directing** NO₂ will **not go** into the **second** position.
 3) **Absence of AlCl₃** and wrong condition for last step.

(ii) Given the first step in the synthesis is as follows:



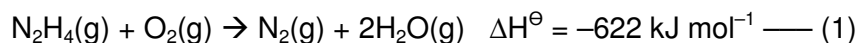
(ii) State the reagents and conditions Steps II and III.

Step II: **KMnO₄ and H⁺, reflux/heat**

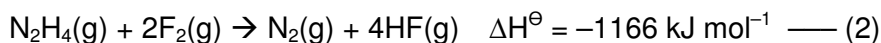
Step III: **Cl₂ in AlCl₃, heat**

[Total:13]

- 3 The major uses of hydrazine, N₂H₄, are in rocket fuels. Hydrazine was first used as a rocket fuel during World War II for the first rocket-powered fighter plane. These reactions are extremely exothermic and they produce large volumes of hot gas from a small volume of liquid hydrazine.



It has been suggested that hydrazine/fluorine mixtures might be more exothermic than hydrazine/oxygen mixtures.



- (a)(i) With the aid of a *Data Booklet*, calculate the enthalpy change per mole of hydrazine when it reacts with fluorine.

Bonds broken: (N–N) + 4(N–H) + 2(F–F)

Bonds formed: (N≡N) + 4(H–F)

$$\begin{aligned} \text{Enthalpy change per mole} &= 160 + 4(390) + 2(158) - 994 - 4(562) \\ &= \mathbf{-1206 \text{ kJ mol}^{-1}} \end{aligned}$$

- (ii) Suggest a reason for the discrepancy between your calculated value in (i) and that given in the question.

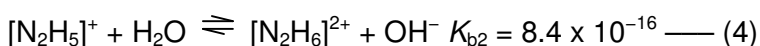
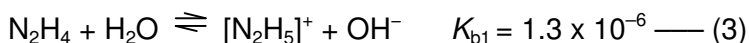
The bond enthalpies given in the data booklet are **average values**.

- (iii) Explain the difference in $\Delta H^\ominus$ between reactions (1) and (2) in terms of bond strength.

Reaction (1) is **less exothermic** than (2) or **more energy is needed to break the O_2 bond** as **bond strength of F–F (158) is lower** than O_2 (496).

OR Reaction (2) is **more exothermic** than (1) or as **more energy is released** from forming **stronger H–F bonds**.

- (b) In some way, hydrazine resembles ammonia in the way it reacts with water. The product from the first reaction can react with a second molecule of water as shown by equation (2)



- (b)(i) Identify one conjugate acid–base pair in equation (1).

N_2H_4 : Base

$[N_2H_5]^+$: Conjugate acid

- (ii) Explain why K_{b2} is much smaller than K_{b1} .

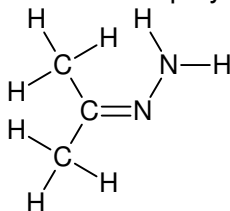
It is **more difficult to accept a proton** from a **positively charged** ion due to charge repulsion.

Hydrazine also find many applications in organic chemistry. One such application is the reaction with carbonyl compound to form hydrazone.

- (c)(i) State the type of reaction which occurred when hydrazine reacts with propanone to form the hydrazone.

Condensation

- (ii) Draw the displayed formula of the organic compound formed.



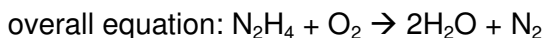
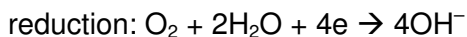
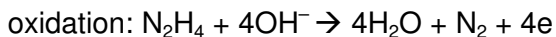
- (iii) Explain why the organic compound formed from the reaction cannot exist as stereoisomers.

Because there are **two same substituents (or $-CH_3$)** attached on the **double bond C**.

- (d) Hydrazine also finds its use in fuel cell to power automobiles and other motorized vehicles. A redox reaction takes place in the cell where hydrazine is oxidised in an

alkaline medium to nitrogen gas while oxygen gas is reduced to hydroxide ions at the cathode where reduction takes place.

Write the half-equations which occurred for the redox reaction of hydrazine, and hence the overall reaction of the fuel cell.

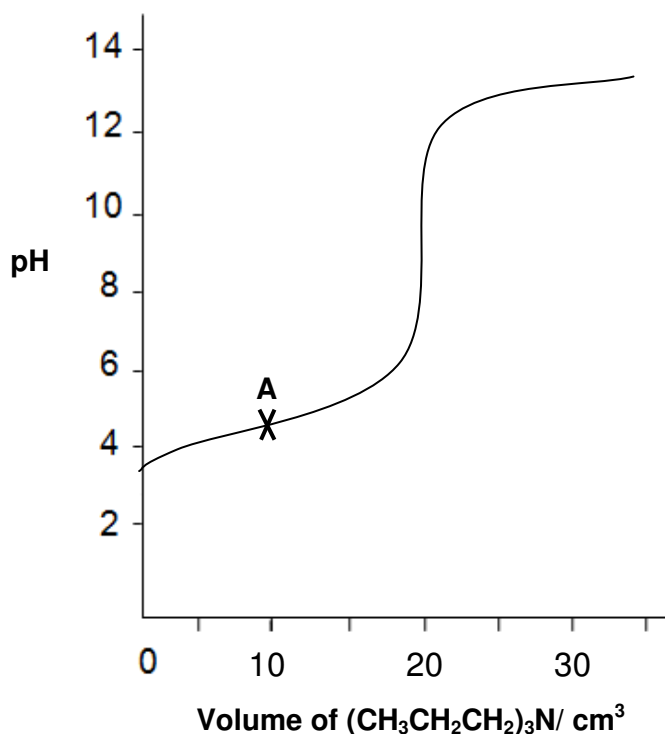


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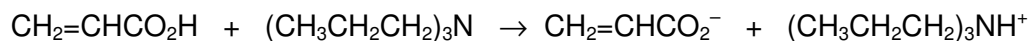
[Total: 12]

Section B

- 1(a) Carboxylic acids are weak acids as they only ionise slightly in water. One example will be the acrylic acid, with the formula of $\text{CH}_2=\text{CHCO}_2\text{H}$. 15.0 cm^3 of 0.30 mol dm^{-3} of acrylic acid is titrated with a strong base, aqueous tripropylamine $(\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{N}$, and its titration curve is as shown below.

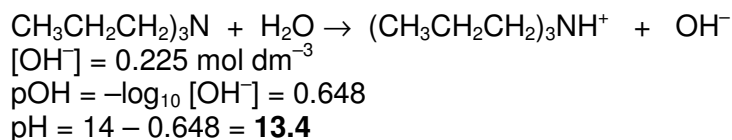


- (i) Write a balanced equation for the reaction between acrylic acid and tripropylamine.



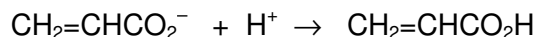
- (ii) Calculate the concentration of tripropylamine used in this titration and hence its pH.

$$\begin{aligned} n(\text{CH}_2=\text{CHCO}_2\text{H}) &= 0.0150 \times 0.30 = 0.0045 \text{ mol} \\ n[(\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{N}] &= 0.0045 \text{ mol} \\ [(\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{N}] &= 0.0045/0.020 = \underline{\underline{0.225 \text{ mol dm}^{-3}}} \end{aligned}$$



- (iii) The solution at point **A** can act as a *buffer*. Explain what is meant by a buffer. Write an equation to show how the solution at point **A** can act as a buffer when a small amount of H^+ is added to it.

A **buffer solution** is one which is capable of **maintaining a fairly constant pH**/resisting pH change when **small amounts** of acid or base are added to it.



- (iv) Suggest with reasoning from the following list, a suitable indicator to be used for the titration.

Indicators	pH range
Methyl violet	0 – 1
Methyl orange	3 – 4
Bromothymol Blue	6 – 7
Phenolphthalein	9 – 10

Phenolphthalein as the **endpoint is > 7**.

- (b) The standard enthalpy change of the above reaction in (a)(i) can be determined experimentally by mixing known volumes of 1.0 mol dm^{-3} acrylic acid and 1.0 mol dm^{-3} tripropylamine. The following results are obtained. [6]

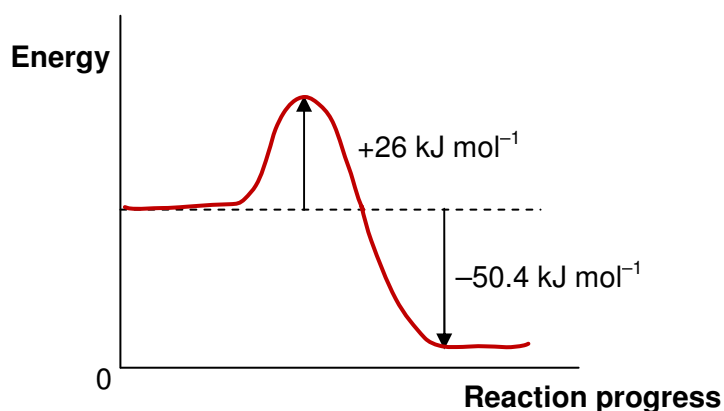
Volume of acrylic acid used = 40.0 cm^3
 Volume of tripropylamine used = 40.0 cm^3
 Initial temperature of mixture = 26°C
 Final temperature of mixture = 32°C

- (i) Use the data given to calculate the standard enthalpy change of the reaction. [Assume specific heat capacity of all solutions = $4.20 \text{ J g}^{-1} \text{ K}^{-1}$].

$$\begin{aligned} Q &= mc\Delta\theta \\ &= (4.20 \times 80.0) \times (26 - 32) \\ &= -2016 \text{ J} \end{aligned}$$

$$\Delta H_{\text{rxn}}^{\theta} = \frac{-2016}{0.040} = \underline{\underline{-50.4 \text{ kJ mol}^{-1}}}$$

- (ii) Using the standard enthalpy change calculated in (i), draw a labelled energy profile diagram for the reaction, given that it has an activation energy of $+26 \text{ kJ mol}^{-1}$.



- (iii) The standard enthalpy change of reaction between the NaOH and HCl is found experimentally to be -57 kJ mol^{-1} . Explain if the ΔH^{θ} calculated in (i) is plausible.

The $\Delta H_{\text{rxn}}^{\theta}$ calculated in b(i) is plausible as a neutralisation reaction between a strong base and weak acid will have a $|\Delta H_{\text{rxn}}^{\theta}| < 57 \text{ kJ mol}^{-1}$ as some amount of energy is required to cause ionisation of the undissociated weak acid molecules in order for neutralisation to occur.

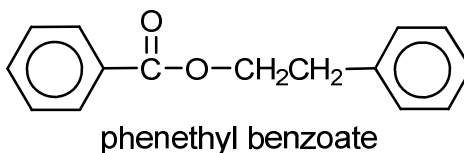
- (iv) State two possible sources of errors in this experiment.

Heat loss to the surrounding and experiment not carried out at standard conditions.

[7]

- (c) A distinct difference between carboxylic acids and esters is in their odors. Carboxylic acids are noted for their sour, disagreeable odors. On the other hand, esters have sweet pleasant odors often associated with fruits. Hence esters are often used in the food industry as flavoring agents.

A food chemist decides to synthesise the following ester, for possible use as a flavouring agent.



The **only** organic compound available is phenylethanal ($\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$). Explain how the food chemist is able to synthesise this ester, stating the necessary reagents and conditions for each step.

1. **Reflux** phenylethanal with KMnO_4/H^+ to get **benzoic acid**.
2. Reduce phenylethanal with LiAlH_4 in **dry ether** to get **2-phenylethanol**, $\text{C}_6\text{H}_5(\text{CH}_2)_2\text{OH}$.
3. **Reflux** benzoic acid and 2-phenylethanol with **conc. H_2SO_4** to get the ester.

[2]

- (d) The label on the test tube containing phenethyl benzoate, which is synthesised in (c), came off and it is mixed up with two other compounds with structures as shown below.

- Phenylmethanol, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- Phenylethene, $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$

Outline a **sequence of chemical tests** by which you could identify each of the above organic substances. You are **not** allowed to identify the substances by elimination.

Step 1: Add $\text{PCl}_3/\text{PCl}_5/\text{SOCl}_2$, r.t

Observations: White fumes of HCl seen $\Rightarrow$ Phenylmethanol
No white fumes $\Rightarrow$ Phenethyl benzoate / Phenylethene

Step 2: Heat/Reflux each of the unknown with dilute NaOH/HCl then add in $\text{PCl}_3/\text{PCl}_5/\text{SOCl}_2$.

Observations: White fumes of HCl seen $\Rightarrow$ Phenethyl benzoate
No white fumes $\Rightarrow$ Phenylethene

Step 3: Add aqueous Br_2 or Br_2 (l) in CCl_4 to phenylethene

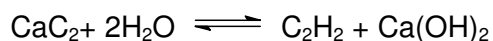
Observations: Decolourisation of reddish-brown aqueous Br_2 or Br_2 (l) in CCl_4
 $\Rightarrow$ Phenylethene is confirmed.

[5]

[Total: 20]

- 2(a)** Calcium carbide, CaC_2 , is an ionic compound of calcium and carbon which is one of the most important carbides used in technology. It is used in carbide lamps, in which water drips on the carbide and the acetylene, C_2H_2 , formed is ignited. Calcium hydroxide is also produced in this *equilibrium* reaction.

- (i) Write an equation to show the production of acetylene from calcium carbide.



- (ii) Draw a dot-and-cross diagram to show the arrangement of electrons in acetylene.



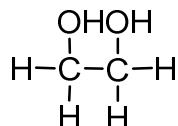
- (iii) State the oxidation number of carbon in acetylene and give an example of a carbon-containing species with the same oxidation number.

$$2x + 2(1) = 0$$

$$x = -1$$

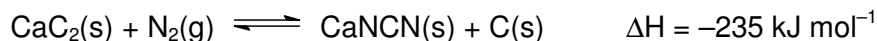
$$\therefore \text{oxidation number of C} = -1$$

$\text{CH}_2(\text{OH})\text{CH}_2(\text{OH})$ has an oxidation number of -1 for its carbon atom.

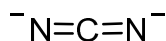


[4]

- (b)** Upon heating calcium carbide in nitrogen at 1000°C , calcium cyanamide is produced:



- (i) State the formula of the cyanamide ion and draw its Lewis structure.



- (ii) Using your answer in (i), predict the shape of cyanamide ion and give the hybridisation of carbon atom.

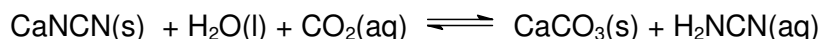
Linear in shape. It is **sp hybridised**.

- (iii) Explain qualitatively why the production of calcium cyanamide is carried out at an optimum temperature of 1000 °C.

According to **Le Chatelier's Principle**, equilibrium will lie to the left at **high temperatures** hence the **yield is low** as the **forward reaction is exothermic**. At **low temperatures**, the **reaction rate will be slow**. An optimum temperature of 1000 °C is required to ensure the reaction proceeds at a reasonably rate with high yield of calcium cyanamide.

[7]

- (c) Calcium cyanamide is often being hydrolysed to give cyanamide, H₂NCN, which is widely used in agriculture and the production of pharmaceuticals and other organic compounds.



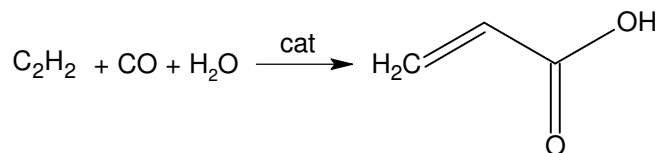
Write an expression for the equilibrium constant, K_c , of the reaction given above. By using the numerical value of this K_c (4.7×10^3) and equilibrium $[\text{CO}_2\text{(aq)}]$ of $3.4 \times 10^{-4} \text{ mol dm}^{-3}$, estimate the concentration of H₂NCN that could be produced.

$$K_c = \frac{[\text{H}_2\text{NCN}]}{[\text{CO}_2]}$$

$$\begin{aligned} [\text{H}_2\text{NCN}] &= K_c[\text{CO}_2] \\ &= (4.7 \times 10^3)(3.4 \times 10^{-4}) \\ &= \underline{\underline{1.60 \text{ mol dm}^{-3}}} \end{aligned}$$

[2]

- (d) Acetylene can react with carbon monoxide and water, in the presence of a catalyst, to give an acrylic acid, which can be used to produce acrylic glass.

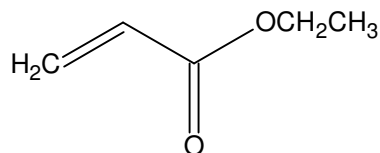


- (i) Compare and explain the relative acidity of this acrylic acid with 2-chloropropanoic acid, CH₃CH₂(Cl)COOH.

2-chloropropanoic acid is **more acidic** than the acrylic acid. The **electron-withdrawing Cl** group **dispersed** the **negative charge** on the **carboxylate ion** hence **stabilising it**.

Acetylene can also react with carbon monoxide and ethanol in a very similar fashion to the above reaction in **part (d)** to give product **A**.

- (ii) Draw the structural formula of **A**.



- (iii) **A** is then subjected to strong oxidation whereby hot acidified KMnO_4 is used in the reaction.

Draw the structural formula of all the products upon strong oxidation of **A**.

CO_2 **and** H_2O
 CH_3COOH

- (iv) Suggest an alternative set of reagent and condition for the oxidation reaction such that there are **only** 2 distinct organic products upon oxidation.

Hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$

[7]

[Total: 20]

3 This question is about phosphorus and its compounds.

- (a) (i) State the full electronic configuration of phosphorus.

$1s^2 2s^2 2p^6 3s^2 3p^3$

- (ii) Suggest how the first ionisation energy of phosphorus would compare to that of sulfur and explain your reasoning.

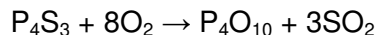
Phosphorus has **higher first ionisation energy** than sulphur.

Inter-electron repulsion between the paired 3p electrons in sulfur makes it **easier to remove** the one of the paired 3p electrons from sulfur than to remove the unpaired 3p electron from phosphorus which does not experience such repulsion.

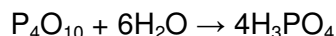
[4]

- (b) One of the products from the reaction of phosphorus and sulfur is phosphorus sulfide, P_4S_3 , which is used in small amounts in the tip of a match. On striking a match, this compound burns to give phosphorus(V) oxide, P_4O_{10} and sulfur dioxide, SO_2 .

- (i) Construct a balanced equation for the reaction of P_4S_3 to form the respective oxides.



- (ii) Both oxides formed in part (b)(i) dissolve in water. Construct an equation for the reaction of each oxide with water.



[3]

- (c) Phosphorus reacts with limited amount of chlorine to give PCl_3 and excess chlorine to form phosphorus pentachloride, PCl_5 .

- (i) Explain why PCl_5 sublimes at 160°C while PCl_3 boils at 76.1°C .

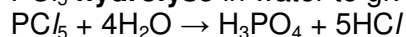
Both PCl_5 and PCl_3 have **simple molecular structures**.

Due to the **larger number of electrons** in the PCl_5 molecule/ **larger electron cloud** of PCl_5 molecule

More energy/ larger amount of energy is required to overcome/break the stronger induced dipole-induced dipole interaction/attraction between PCl_5 molecules than the weaker permanent dipole-permanent dipole interaction/attraction between PCl_3 molecules.

- (ii) Describe, with an equation, the reaction of PCl_5 , with water. What would be the effect of adding universal indicator to the product?

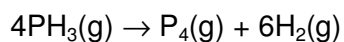
PCl_5 **hydrolyse** in water to give **acidic** solutions.



The universal indicator turned from **green to red**.

[6]

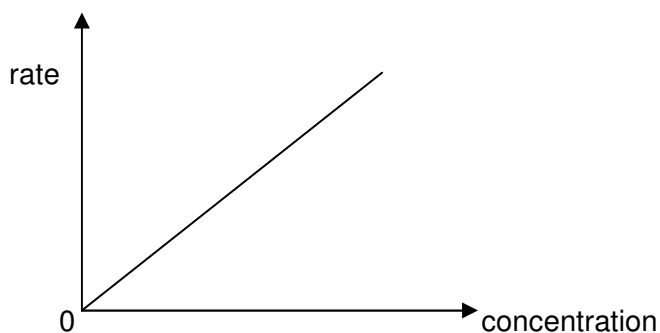
- (d) Phosphine, PH_3 , is a widely used fumigant. The thermal decomposition of phosphine into phosphorus and hydrogen is a *first-order reaction*. The half-life of the reaction is 35.0 seconds at 680 °C.



- (i) Explain the term in *italics*.

A first-order reaction refers to one in which the **order of reaction with respect to the reactant** is **one** in the rate equation.

- (ii) Sketch a rate against concentration for the decomposition reaction.



- (iii) Calculate the rate constant for the reaction.

$$k = \ln 2 / t_{1/2} \\ = \ln 2 / 35.0 = \underline{\underline{0.0198 \text{ s}^{-1}}}$$

- (iv) Calculate the time taken for 87.5% of the phosphine to decompose.

$$\text{Time taken} = 3 \times 35.0 = \underline{\underline{105 \text{ s}}}$$

[7]
[Total: 20]